Table 7.3: Enzymes of *Streptococcus pyogenes* and its virulence nature

Enzymes	Virulence nature
streptokinase (fibrinolysin)	It promotes the lysis of human fibrin clot by catalyzing the conversion of plasminogen into plasmin. It facilitates the spread of infection by breaking down the fibrin barrier around the lesions.
Deoxyribonucleases	It liquefy the highly viscous DNA that accumulate in thick pusand responsible for thin serous character of streptococcal exudates
Hyaluronidase	It breaks down hyaluronic acid of the tissues and favors spread of streptococcal lesion along intercellular spaces.
Other enzymes	NADase, lipase, amylase, esterase, phosphates and other enzymes

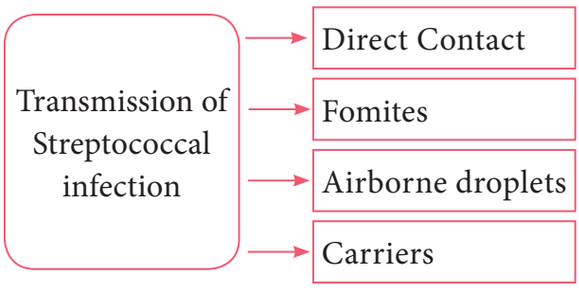


Streptokinase: It is given intravenously for the treatment of early myocardial infarction and other thromboembolic disorders. *Streptococcus equisimilis* is the source of streptokinase used for thrombolytic therapy in patients.

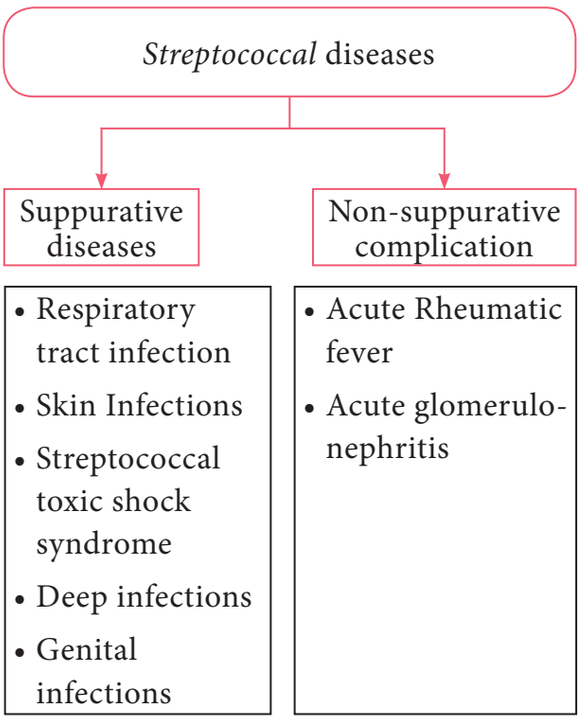
7.4.4 Pathogenesis

Streptococcus pyogenes is intrinsically a much more dangerous pathogen than *Staphylococcus aureus* and has a much greater tendency to spread in the tissues.

Mode of transmission: Streptococcal infections are transmitted by the following ways:



Streptococcal diseases may be broadly classified, and it is shown in flowchart 7.1



Flowchart 7.1: Classification of *Streptococcal* diseases

Suppurative Infections

1. Respiratory tract infection

a. Streptococcal sore throat: Sore throat (acute tonsillitis and pharyngitis) is the

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